

Code Explanation

Data Loader Module Explanation

The provided code is a data loader module written in Python, utilizing the PySpark library to load data from Oracle databases. It is designed to handle the loading of source and lookup tables, and then enrich the source data with the lookup information.

Overview

This data loader module is responsible for reading data from Oracle source and lookup tables using PySpark. It provides methods to load staging data and lookup data from specific tables, and then enrich the staging data with the loaded lookup information.

Step 1: Initialization of DataLoader Class

The DataLoader class is initialized with a SparkSession object, which is used to create DataFrames and interact with the Oracle databases. The class also loads configuration settings from a Config object.

```
def __init__(self, spark: SparkSession):  
    self.spark = spark  
    self.config = Config()
```

Step 2: Loading Staging Data from Oracle

The `load\_staging\_data` method loads data from the STG_E2E_AC_TXDBH_DATA table in the Oracle database. It uses the `spark.read.jdbc` method to read the data and then selects the required columns based on the S2T mapping.

```
df = self.spark.read.jdbc(  
    url=self.config.get_oracle_jdbc_url(),  
    table=self.config.SOURCE_TABLE,  
    properties=self.config.get_oracle_properties()  
)  
required_cols = [  
    "T_TX_REQUEST_REQUEST_ID",  
    "T_TX_BASIC_TRANS_ID",  
    # ... other required columns  
)  
return df.select(*[col(c) for c in required_cols if c in df.columns])
```

Step 3: Loading Lookup Data from Oracle

The `load\_lookup\_yuyu\_clnt` and `load\_lookup\_yuyuk\_cln` methods load lookup data from the T_YUYU_CLNT and T_YUYUK_CLN tables, respectively. They use the same `spark.read.jdbc` method as in Step 2 and select the required columns.

```
df = self.spark.read.jdbc(  
    url=self.config.get_oracle_jdbc_url(),  
    table=self.config.LOOKUP_TABLE_YUYU_CLNT,  
    properties=self.config.get_oracle_properties()  
)  
return df.select(  

```

```
col("POL_NO").alias("LKP_POL_NO_CLNT"),  
col("POWN_LNM"),  
col("POWN_FNM")  
)
```

Step 4: Enriching Staging Data with Lookup Information

The `enrich\_with\_lookups` method enriches the staging data with the loaded lookup information. It performs left joins with the T_YUYU_CLNT and T_YUYUK_CLN DataFrames based on the policy ID.

```
enriched_df = source_df.join(  
    lookup_clnt_df,  
    source_df.T_TX_BASIC_POLICY_ID == lookup_clnt_df.LKP_POL_NO_CLNT,  
    "left"  
).drop("LKP_POL_NO_CLNT")  
enriched_df = enriched_df.join(  
    lookup_cln_df,  
    source_df.T_TX_BASIC_POLICY_ID == lookup_cln_df.LKP_POL_NO_CLN,  
    "left"  
).drop("LKP_POL_NO_CLN")  
return enriched_df
```